

PYTHON PATTERN PROBLEMS (09-07-26)

#1

```
rows = 6
for i in range(rows):
    for j in range(8):
        if (
            (i == 0 and j == 0) or
            (i == 1 and (j == 0 or j == 2)) or
            (i == 2 and (j == 0 or j == 4)) or
            (i == 3 and j < 5) or
            (i == 4 and (j == 0 or j == 6)) or
            (i == 5 and (j == 0 or j == 7))
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#2

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            j == 0 or
            (i == 0 and j < 4) or
            (i == 2 and j < 4) or
            (i == 4 and j < 4) or
            (j == 4 and (i == 1 or i == 3))
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#3

```
rows = 5
cols = 5

for i in range(rows):
    for j in range(cols):
        if i == 0 or i == rows - 1 or j == 0:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#4

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            j == 0 or
            (i == 0 and j < cols - 1) or
            (i == rows - 1 and j < cols - 1) or
            (j == cols - 1 and i > 0 and i < rows - 1)
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#5

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            j == 0 or
            i == 0 or
            i == 2 or
            i == 4
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#6

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            j == 0 or
            i == 0 or
            i == 2
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#7

```
rows = 5
cols = 6
for i in range(rows):
    for j in range(cols):
        if (
            i == 0 or
            i == 4 or
            j == 0 or
            (i == 2 and j >= 2) or
            (j == 5 and i >= 2)
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#8

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            j == 0 or
            j == cols - 1 or
            i == 2
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#9

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            i == 0 or
            i == rows - 1 or
            j == cols // 2
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#10

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            (j == 4 and i < 4) or
            (j == 0 and i == 3) or
            (i == 4 and j > 0 and j < 4)
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#11

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            j == 0 or
            (i + j == 2) or
            (i - j == 2)
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#12

```
rows = 5
cols = 5

for i in range(rows):
    for j in range(cols):
        if (
            j == 0 or
            i == rows - 1
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#13

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            j == 0 or
            j == cols - 1 or
            (i == 1 and (j == 1 or j == 3)) or
            (i == 2 and j == 2)
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#14

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            j == 0 or
            j == cols - 1 or
            i == j
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#15

```
rows = 5
cols = 5

for i in range(rows):
    for j in range(cols):
        if (
            ((i == 0 or i == rows - 1) and j > 0 and j < cols - 1) or
            ((j == 0 or j == cols - 1) and i > 0 and i < rows - 1)
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#16

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            j == 0 or
            (i == 0 and j < cols - 1) or
            (i == 2 and j < cols - 1) or
            (j == cols - 1 and i == 1)
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#17

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            (i == 0 and j > 0 and j < 4) or
            (j == 0 and i > 0 and i < 3) or
            (j == 4 and i > 0 and i < 3) or
            (i == 3 and j == 1) or
            (i == 3 and j == 4) or
            (i == 4 and j == 2) or
            (i == 4 and j == 3)
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#18

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            j == 0 or
            (i == 0 and j < cols - 1) or
            (i == 2 and j < cols - 1) or
            (j == cols - 1 and i == 1) or
            (i == 3 and j == 2) or
            (i == 4 and j == 3)
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#19

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            (i == 0 and j > 0) or
            (j == 0 and i == 1) or
            (i == 2 and j > 0) or
            (j == cols - 1 and i == 3) or
            (i == 4 and j < cols - 1)
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#20

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            i == 0 or
            j == cols // 2
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#21

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            ((j == 0 or j == cols - 1) and i < rows - 1) or
            (i == rows - 1 and j > 0 and j < cols - 1)
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#22

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            (i < 2 and (j == 0 or j == cols - 1)) or
            (i == 2 and (j == 1 or j == 3)) or
            (i >= 3 and j == cols // 2)
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#23

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            j == 0 or
            j == cols - 1 or
            (i == 2 and j == 2) or
            (i == 3 and (j == 1 or j == 3))
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```


#24

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            i == j or
            i + j == cols - 1
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#25

```
rows = 5
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            (i == 0 and (j == 0 or j == 4)) or
            (i == 1 and (j == 1 or j == 3)) or
            (j == 2 and i >= 2)
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

#26

```
rows = 6
cols = 5
for i in range(rows):
    for j in range(cols):
        if (
            i == 0 or
            i == rows - 1 or
            i + j == cols - 1
        ):
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

